

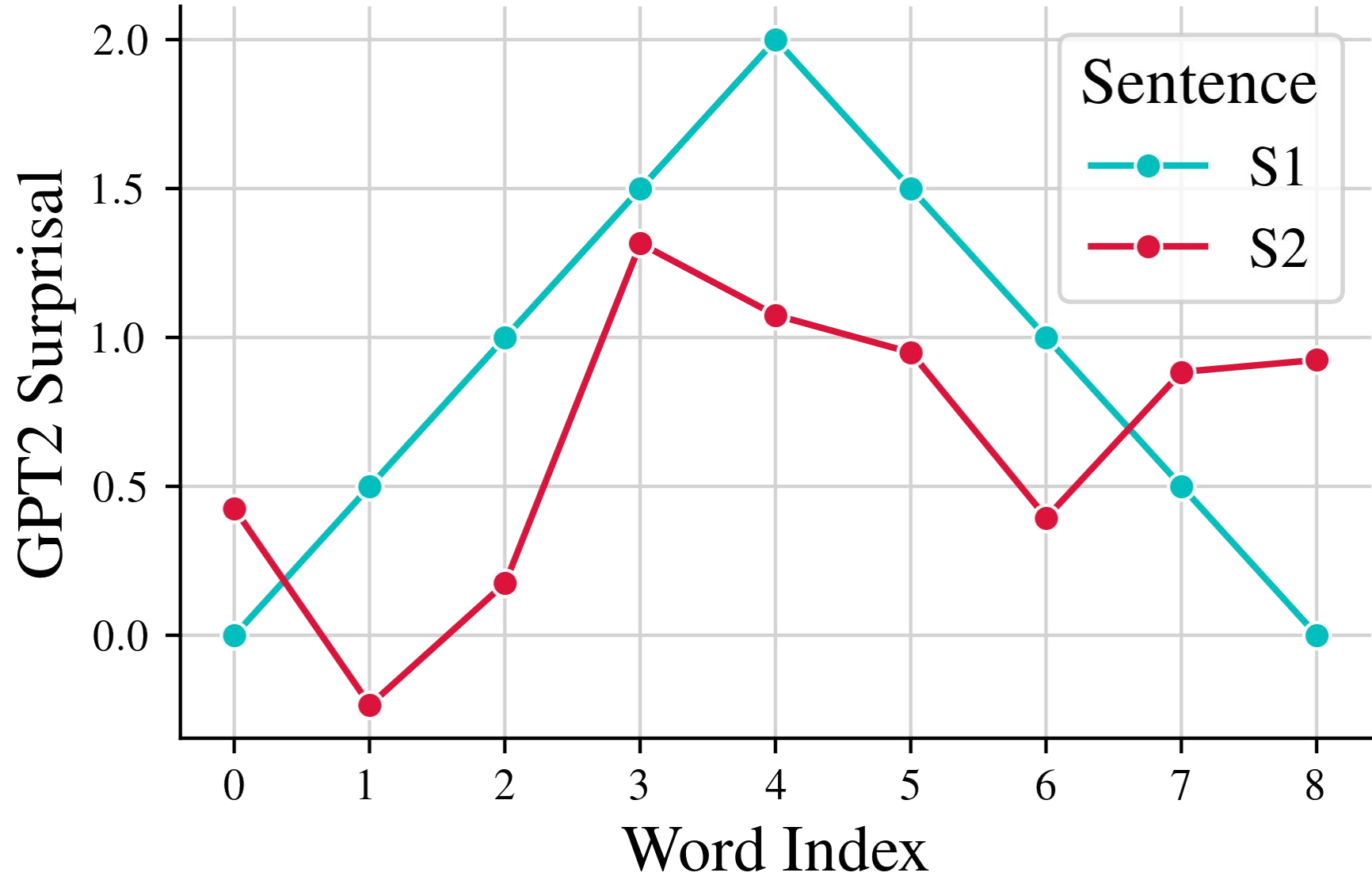
An Information-Theoretic Study of RLHF-Induced Uniformity in Large Language Model Outputs

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Background



LLMs now write as many online articles as humans, but outputs still differ in many ways
Humans are theorized to prefer Uniform Information Density
- Prior work: pretrained LLMs *too uniform*

How does training on human preferences change the uniformity of LLM outputs?

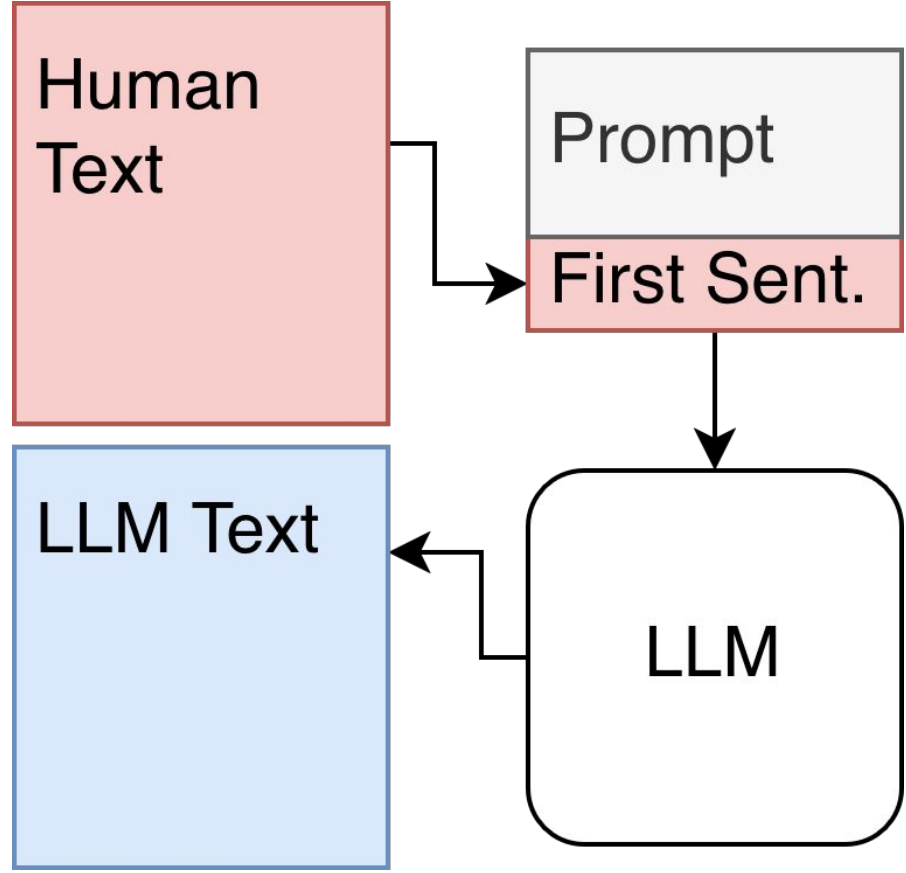
Hypotheses

Hypothesis	Prediction
1) Human-like uniformity is preferable to humans	RLHF reduces uniformity
2) Lower uniformity is more interesting to humans	RLHF reduces uniformity
3) Higher uniformity is easier to comprehend	RLHF increases uniformity
4) RLHF doesn't capture preference for uniformity; uniformity outranked by other factors	RLHF does not change uniformity

Methods

Dataset Generation

1. Sampled human-authored texts from 4 datasets
2. Prompted LLMs to recreate texts from human texts
3. Resulted in 12,000 pairs of human/LLM texts



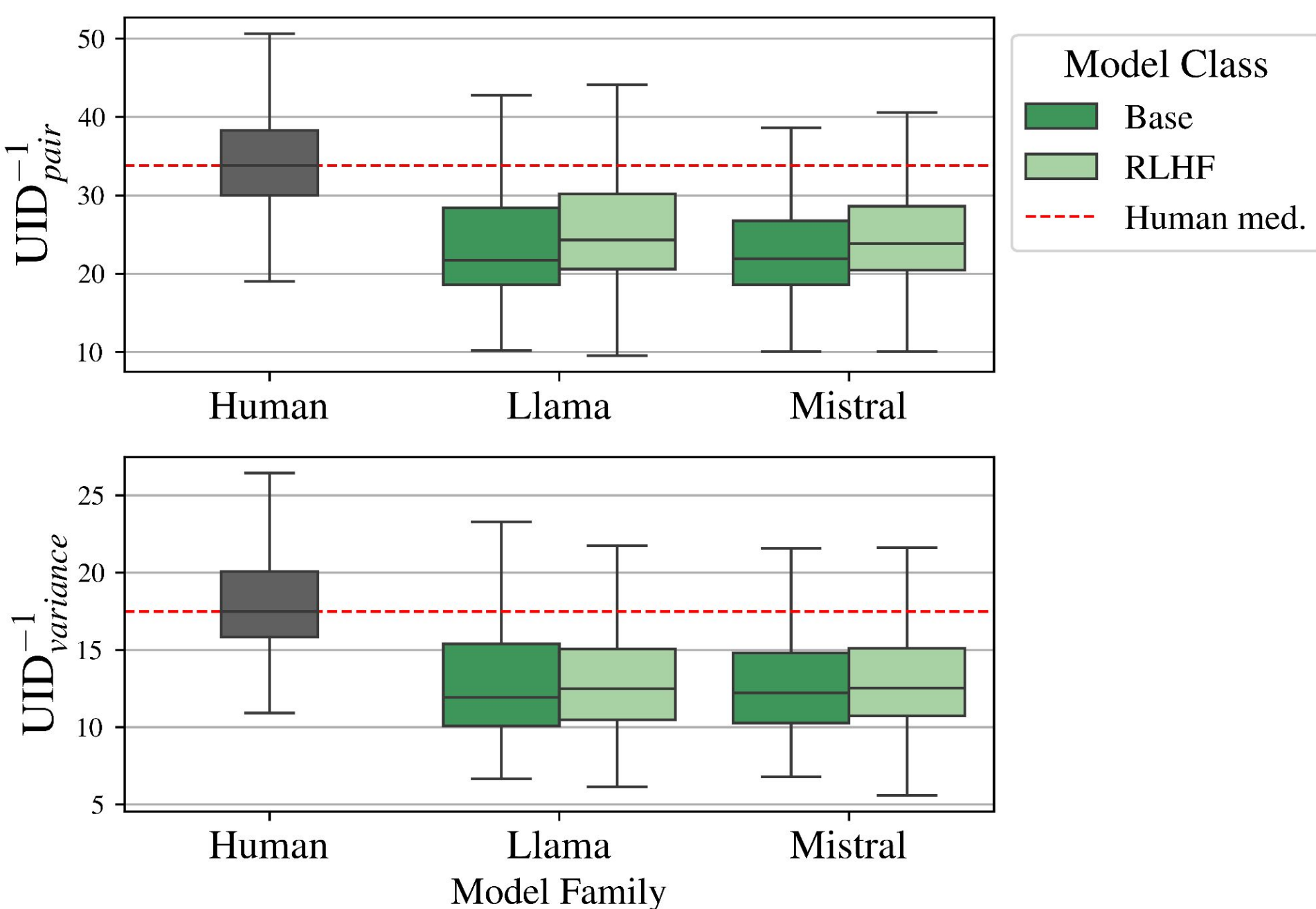
UID Calculation

1. GPT2 estimates token probabilities for surprisal & information contour
2. Calculate uniformity metrics: Surprisal Variance and Local Variance
 - a. SV: Adherence to global mean
 - b. LV: Smooth local transitions

$$UID_{variance}^{-1}(\vec{w}) = \frac{1}{|\vec{w}|} \sum_{i=2}^{|\vec{w}|} (I(w_i) - \mu)^2$$

$$UID_{pair}^{-1}(\vec{w}) = \frac{1}{|\vec{w}|} \sum_{i=2}^{|\vec{w}|} (I(w_i) - I(w_{i-1}))^2$$

RLHF Experiment Results

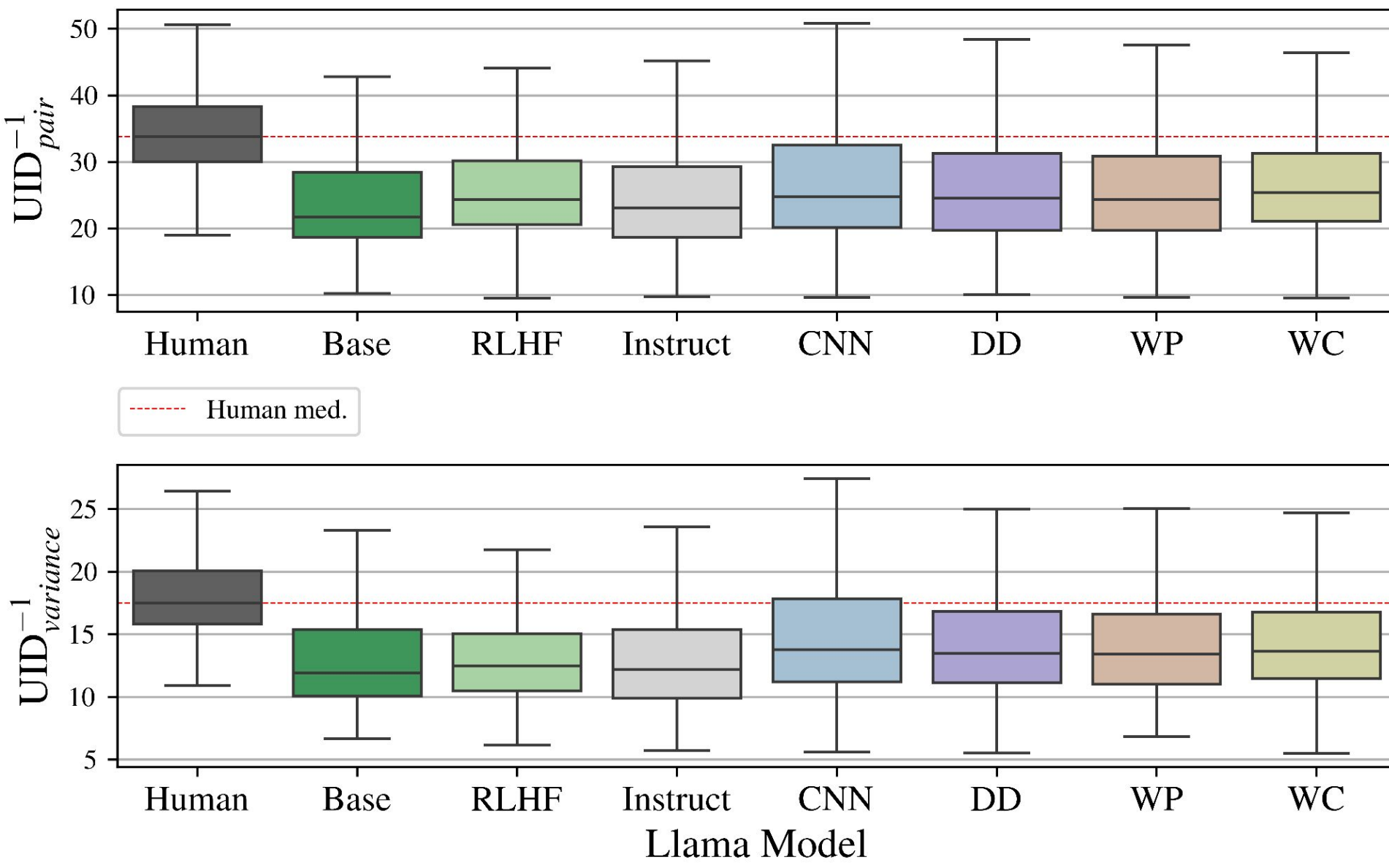


RLHF *slightly reduces* median uniformity
- Consistent across families/datasets, but not significant ($p>0.05$)
- Supports hypotheses 1 and 2, but weakly

STD of...	Human	Base	RLHF
UID_{pair}^{-1}	9.03	13.45	11.49
$UID_{variance}^{-1}$	4.79	6.49	5.14

RLHF *significantly reduces variance* in uniformity
- Texts more consistent in information patterns ($p<0.05$)
- Moves in direction of human texts

Domain Adaptation Results



Domain adaptation *also reduces uniformity*
- Effect is *not* specific to RLHF/human preference training
- Also not significant change ($p<0.05$)
- Supports hypothesis 4

Llama Model	ΔIQR for UID^{-1}	p
RLHF	-0.743	0.007
Instruct	+0.185	0.519
CNN/DailyMail	+1.304	0.001
DailyDialog	+0.368	0.305
WritingPrompts	+0.318	0.363
WildChat	-0.020	0.957

No significant reduction to variance in uniformity from domain adaptation
- For some datasets, variance significantly increases
- Suggests that this effect is unique to RLHF

Conclusion

The process of training on human preferences seems to induce a finer level of control over information flow, decreasing the variance in uniformity across different generated texts.

Human preferences have no effect on median uniformity

Changes in center were not significant
Shifts were reproduced with domain adaptation
- Supports hypothesis 4: other considerations that outweigh uniformity, or RLHF is unable to capture this preference

RLHF induces more consistency in uniformity

Significant decreases in spread of uniformity
RLHF models produced more consistent texts
- Aligned better with human texts
Not reproducible by domain adaptation

Change in overall surprisal varied by family

SFT decreased for both families
RLHF increased for Mistral, but decreased for Llama

Models were still much less surprising and more uniform than humans

Even after RLHF/SFT, models still significantly different
- Model/human differences were much larger than model/model differences

Limitations

Model Prompting

Prompting strategy minimized intervention
- No explicit instruction vis-a-vis UID
Different prompts may yield different results

UID Calculation

Used GPT2 for surprisal estimation based on reading-time scores
Validated results with Qwen2.5 - different magnitudes, same results

Language

Study was done fully in English
Language models in different languages may produce different information patterns

Acknowledgements

We would like to thank Dr. Mario Giulianelli for his insights during the formulation of this paper.

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